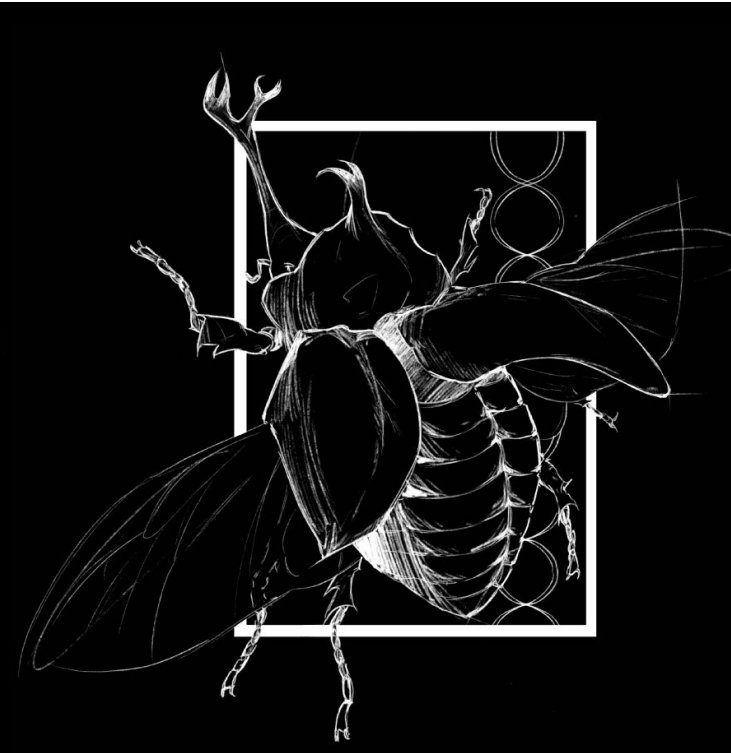




Background

Genome size varies widely across plant lineages and plays a fundamental role in shaping cellular and physiological traits. Differences in genome size are associated with variation in cell size, growth rates, and metabolic demands, which can influence how plants interact with their environment. As a result, genome size variation may affect a species' ecological niche breadth, the range of environments it can tolerate and occupy. In highly diverse plant families such as the Asteraceae, this variation may contribute to differences in ecological strategies and the ability of species to occupy different habitats.

One proposed outcome of genome size variation is its potential to constrain environmental flexibility. Larger genomes have been hypothesized to impose physiological limitations that reduce a species' ability to occupy a wide range of habitats. However, the relationship between genome size and ecological niche breadth remains poorly understood. This research project test whether genome size is associated with climate niche breadth across the Asteraceae.



Model Fitting

climate niche breadth was quantified using both univariate and multivariate approaches derived from species occurrence data. Temperature and precipitation breadth were calculated as the range (maximum minus minimum) of bioclimatic variables associated with each species' geographic records. These measures provided simple estimates of environmental tolerance along individual climate axes.

To capture overall environmental variation, principal componenet analysis (PCA) was applied to mutiple bioclimatic variables. Multivariate niche breadth was then defined as the spread of species occurrences across the first three principal components, representing major axes of climatic variation. This approach allowed ecological breadth to be quantified in a reduced, multidimensional evironemantal space.

Relationships between genome size (2C DNA) and niche breadth were evaluated using correlation analyses and phylogenetic generalized least squares (PGLS). PGLS models accounted for shared evolutionary history among species using a pruned phylogeny, and Pagel's λ was estimated to quantify phylogenetic signal. This framework enabled testing of whether genome size predicts ecological breadth while controlling for phylogenetic relatedness.

Study Design & AI

Genome size data, species occurrence records, and global climate variables were integrated to quantify ecological niche breadth across Asteraceae species. Occurrence data from GBIF were paired with bioclimatic variables to derive univariate and multivariate measures of environmental breadth. Analyses were conducted in R using a reporducible pipeline of script for data processing, statistical modeling, and visualization. Artificial Intelligence (ChatGPT) assisted with R script construction, code debugging,and drafting analyses. This assistance allowed interpretations and all final decisions to be made completely independently.

Phylogenetic information was incorporated to account for shared evolutionary history, allowing relationships between genome size and ecological niche breadth to be evaluated within an evolutionary framework.



Results

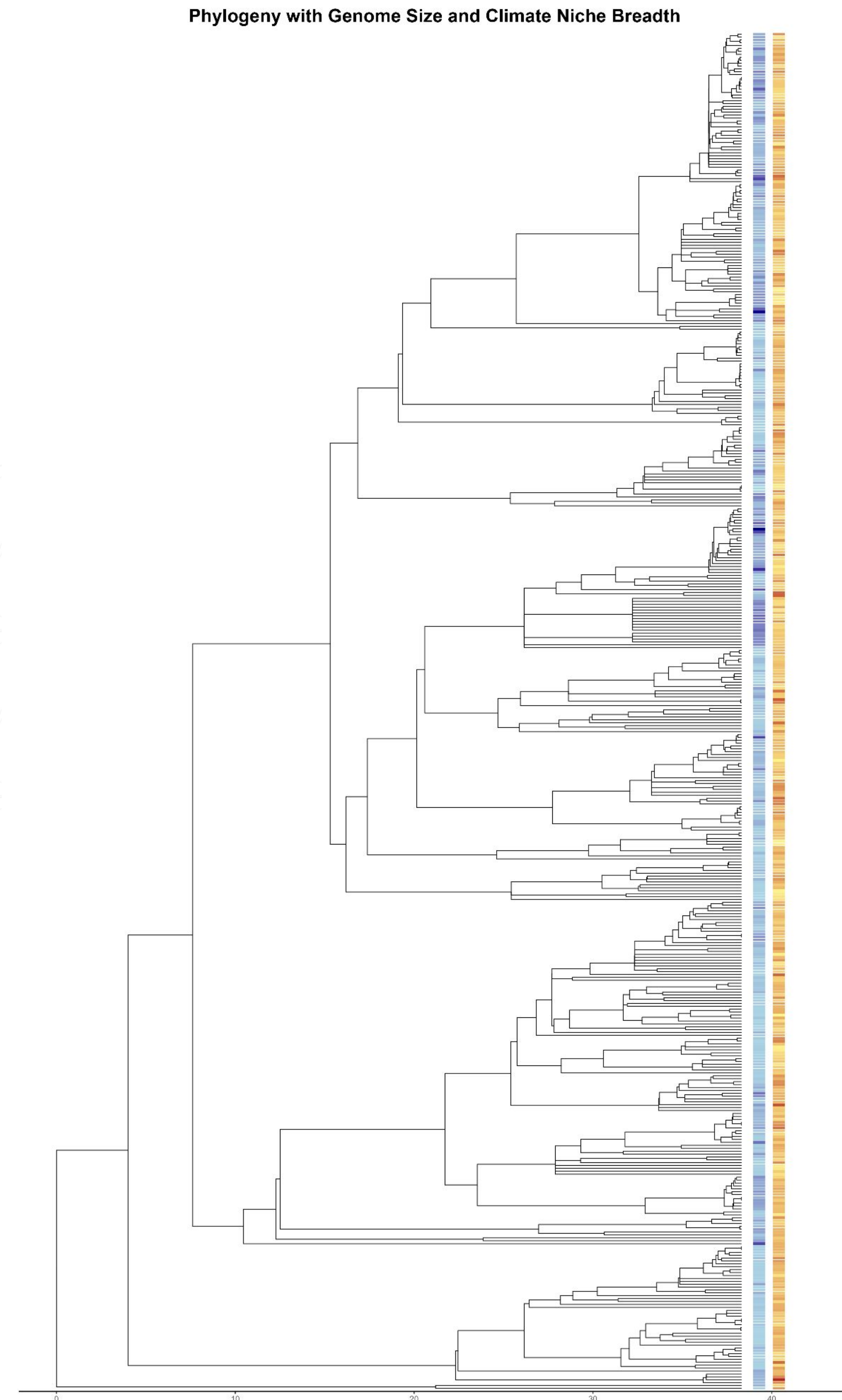


Figure 1. Phylogeny with genome size and niche breadth (n = 470). Phylogenetic tree of 470 Asteraceae species with genome size (blue gradient) and climate niche breadth (red gradient) mapped onto the tips. Trait distributions highlight variation across lineages and reveal strong phylogenetic signal in genome size.

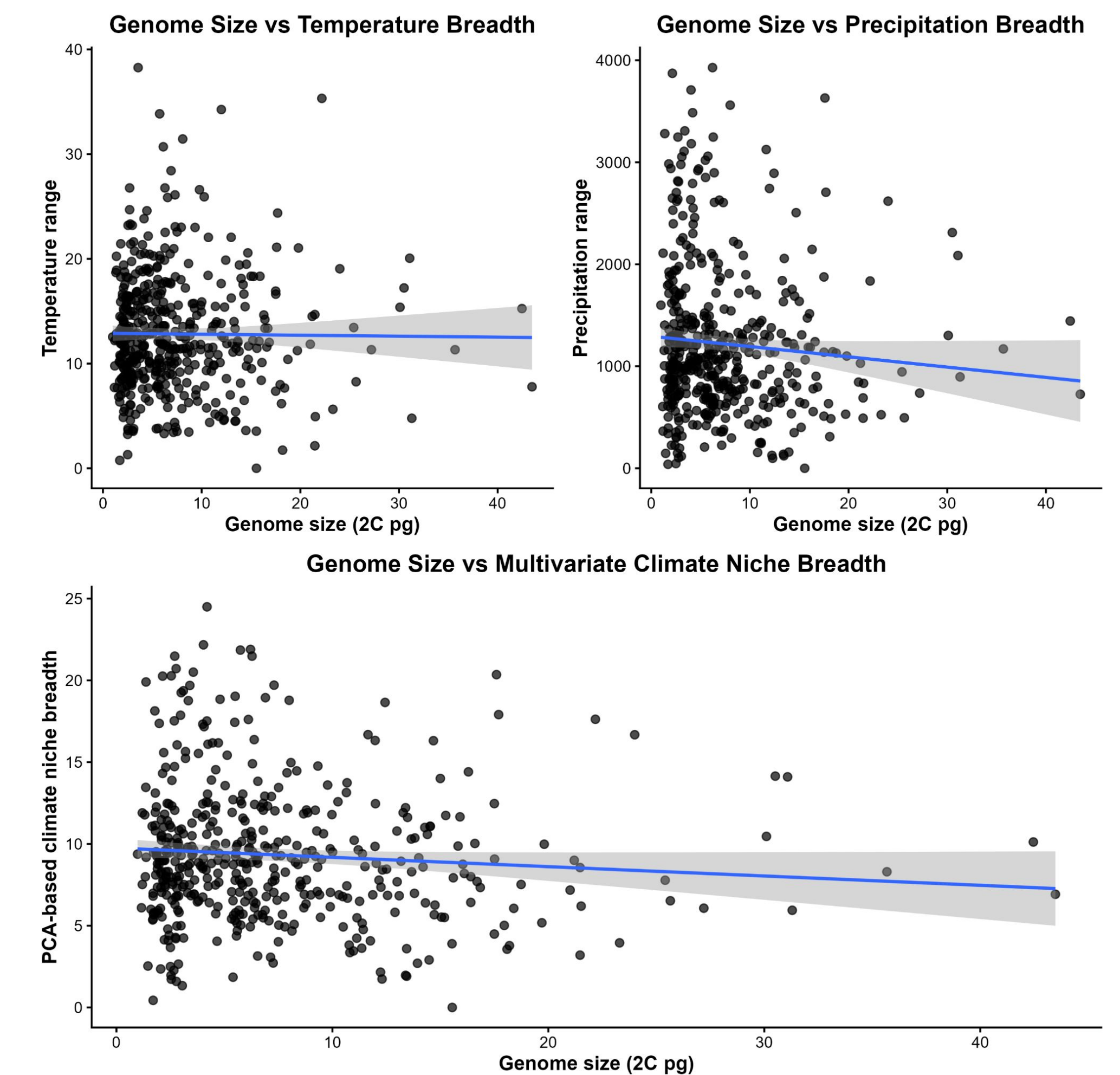


Figure 2. Relationship between genome size and climate niche breadth. Combined panel showing relationship between genome size and temperature breadth ($r = -0.034$, $p = 0.461$), precipitation breadth ($r = -0.087$, $p = 0.060$), and multivariate niche breadth ($r = -0.075$, $p = 0.106$) derived from PCA. Each point represents a species, and regression lines indicate weak, non-significant relationships across all measures of environmental breadth.

Genome Size showed no significant relationship with univariate measures of climate niche breadth across the 470 Asteraceae species. Weak negative correlations were observed between genome size and temperature range ($r = -0.034$, $p = 0.461$; Fig. 2) and precipitation range ($r = -0.087$, $p = 0.060$; Fig. 2), indicating that larger genomes were not consistently associated with reduced environmental breadth.

Similarly, no significant relationship was detected between genome size and multivariate niche breadth derived from principal components analysis ($r = -0.075$, $p = 0.106$; Fig. S4). This suggests that genome size does not strongly predict overall environmentla tolerance when multiple climatic variables ae considered simultaneously.

Phylogenetic generalized least squares (PGLS) analyses accounting for shared evolutionary history also indicated no significant effect of genome size on nich ebreadth ($p = 0.309$; Fig. S4). However, genome size exhibited strong phylogenetic signal (Pagel's $\lambda \approx 0.69$, $p < 0.001$), indicating that closely related species tend to have similar genome sizes (Fig. 1).

Overall, these results indicate that while genome size is evolutionarily conserved, it is not a strong predictor of ecological niche breadth across Asteraceae.

Discussion

Across Asteraceae species, genome size showed no significant relationship with either univariate measures of climate breadth (temperature and precipitation range) or multivariate niche breadth derived from PCA. Although weak negative trends were observed, these relationships were not statistically significant after accounting for phylogenetic relatedness, suggesting that genome size alone does not strongly constrain ecological niche breadth at this scale.

Phylogenetic analyses revealed a strong signal in genome size, indicating that closely related species tend to have similar genome sizes. This pattern suggests that genome size is evolutionary conserved, which may limit its independent influence on ecological breadth. Instead, ecological flexibility is likely shaped by a combination of traits and environmental interactions rather than genome size alone.

These findings highlight the complexity of trait environment relationships and suggest that genome size may play a more indirect or context dependent role in shaping ecological strategies. Future research could incorporate additional functional traits, finer-scale enviromnetal variables, or lineage specific analyses to better understand how genome size interacts with ecological and evolutionary processes.

Conclusions & Acknowledgements

Genome size was not a significant predictor of climate niche breadth across Asteraceae, despite weak negative trends observed in both univariate and multivariate analyses. These findings suggest that genome size alone does not strongly constrain ecological flexibility across this diverse plant family.

In contrast, genome size exhibited strong phylogenetic signal ($\lambda \approx 0.69$), indicating that it is evolutionary conserved among closely related species. This suggests that while genome size reflects shared evolutionary history, ecological niche breadth is likely shaped by a combination of additional traits and environmental interactions.

Overall, these results highlight the complexity of trait environment relationships and suggest that ecological breadth cannot be explained by genome size alone. Future work incorporating additional functional traits and finer-scale environmental data may provide further insight into the mechanisms underlying ecological diversification.

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